

Parametric form of a function or relation — Solutions

Warm-up

Question 1:

To find the Cartesian equation, we eliminate the parameter t .

From the equation for x :

$$\begin{aligned}x &= 2t \\ \implies t &= \frac{x}{2}\end{aligned}$$

Substitute this expression for t into the equation for y :

$$\begin{aligned}y &= 3t - 1 \\ y &= 3\left(\frac{x}{2}\right) - 1 \\ y &= \frac{3}{2}x - 1\end{aligned}$$

The Cartesian equation is $y = \frac{3}{2}x - 1$.

Question 2:

We eliminate the parameter t by solving for t in the first equation.

$$\begin{aligned}x &= t + 1 \\ \implies t &= x - 1\end{aligned}$$

Now, substitute this into the equation for y :

$$\begin{aligned}y &= t^2 \\ y &= (x - 1)^2\end{aligned}$$

Expanding the square gives the Cartesian equation:

$$y = x^2 - 2x + 1$$

Alternatively, $y = (x - 1)^2$ is also a valid Cartesian form.

Question 3:

We eliminate the parameter t by solving for t in the first equation.

$$\begin{aligned}x &= 3t \\ \implies t &= \frac{x}{3}\end{aligned}$$

Substitute this into the equation for y :

$$y = \frac{2}{t}$$

$$y = \frac{2}{x/3}$$

$$y = \frac{6}{x}$$

Thus, the Cartesian equation is $y = \frac{6}{x}$ (where $x \neq 0$).

Question 4:

Given the parametric equations $x = \cos \theta$ and $y = \sin \theta$, we use the fundamental trigonometric identity $\sin^2 \theta + \cos^2 \theta = 1$.

$$\cos \theta = x$$

$$\sin \theta = y$$

Substituting these into the identity:

$$(x)^2 + (y)^2 = 1$$

$$x^2 + y^2 = 1$$

This is the equation of a circle centered at the origin with radius 1.

Question 5:

To find the coordinates of the point on the curve where $t = 3$, we substitute $t = 3$ into both parametric equations.

$$x = t^2$$

$$x = 3^2 = 9$$

$$y = 2t$$

$$y = 2(3) = 6$$

Therefore, the coordinates of the point are $(9, 6)$.

Question 6:

We express y in terms of x by eliminating the parameter t .

From the equation for x :

$$x = 4 - t$$

$$\implies t = 4 - x$$

Substitute this into the equation for y :

$$y = 2t + 5$$

$$y = 2(4 - x) + 5$$

$$y = 8 - 2x + 5$$

$$y = -2x + 13$$

Thus, $y = -2x + 13$.

Question 7:

We use the trigonometric identity $\sin^2 \theta + \cos^2 \theta = 1$.

$$\begin{aligned}\sin \theta &= \frac{x}{2} \\ \cos \theta &= \frac{y}{2}\end{aligned}$$

Substituting these into the identity:

$$\begin{aligned}\left(\frac{x}{2}\right)^2 + \left(\frac{y}{2}\right)^2 &= 1 \\ \frac{x^2}{4} + \frac{y^2}{4} &= 1 \\ x^2 + y^2 &= 4\end{aligned}$$

The Cartesian equation is $x^2 + y^2 = 4$.

Question 8:

We are given the point $(2, 4)$ and the parametric equations $x = t$ and $y = t^2$.

From the x -coordinate:

$$\begin{aligned}x &= t \\ 2 &= t\end{aligned}$$

Checking this with the y -coordinate:

$$\begin{aligned}y &= t^2 \\ 4 &= 2^2 \\ 4 &= 4 \quad (\text{Consistent})\end{aligned}$$

Thus, the value of the parameter is $t = 2$.

Question 9:

We eliminate the parameter t from the equations $x = 2^t$ and $y = 2^{2t}$.

$$\begin{aligned}y &= 2^{2t} \\ y &= (2^t)^2\end{aligned}$$

Since $x = 2^t$, we substitute x into the equation for y :

$$y = x^2$$

Note that since $x = 2^t$, x must be greater than 0. The Cartesian equation is $y = x^2, x > 0$.

Question 10:

We eliminate the parameter t by solving for t in the first equation.

$$x = t - 3$$

$$\implies t = x + 3$$

Substitute this into the equation for y :

$$y = 3t$$

$$y = 3(x + 3)$$

$$y = 3x + 9$$

The Cartesian equation is $y = 3x + 9$.

Skill-building

Question 1:

We eliminate the parameter θ using the identity $\sin^2 \theta + \cos^2 \theta = 1$.

$$\cos \theta = \frac{x + 1}{3}$$

$$\sin \theta = \frac{y - 2}{3}$$

Substituting these into the identity:

$$\left(\frac{x + 1}{3}\right)^2 + \left(\frac{y - 2}{3}\right)^2 = 1$$

$$\frac{(x + 1)^2}{9} + \frac{(y - 2)^2}{9} = 1$$

$$(x + 1)^2 + (y - 2)^2 = 9$$

This is the equation of a circle with centre $(-1, 2)$ and radius 3. The geometric shape is a circle.

Question 2:

We find the intersection of the curve and the line $y = x + 1$ by substituting the parametric equations into the line equation.

$$(2t + 1) = (t^2 - 2) + 1$$

$$2t + 1 = t^2 - 1$$

$$0 = t^2 - 2t - 2$$

Solving for t using the quadratic formula $t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

$$t = \frac{2 \pm \sqrt{(-2)^2 - 4(1)(-2)}}{2(1)}$$

$$t = \frac{2 \pm \sqrt{4+8}}{2}$$

$$t = \frac{2 \pm \sqrt{12}}{2}$$

$$t = \frac{2 \pm 2\sqrt{3}}{2}$$

$$t = 1 \pm \sqrt{3}$$

Now find the coordinates for each t : For $t = 1 + \sqrt{3}$:

$$x = (1 + \sqrt{3})^2 - 2 = 1 + 2\sqrt{3} + 3 - 2 = 2 + 2\sqrt{3}$$

$$y = 2(1 + \sqrt{3}) + 1 = 3 + 2\sqrt{3}$$

For $t = 1 - \sqrt{3}$:

$$x = (1 - \sqrt{3})^2 - 2 = 1 - 2\sqrt{3} + 3 - 2 = 2 - 2\sqrt{3}$$

$$y = 2(1 - \sqrt{3}) + 1 = 3 - 2\sqrt{3}$$

Points of intersection are $(2 + 2\sqrt{3}, 3 + 2\sqrt{3})$ and $(2 - 2\sqrt{3}, 3 - 2\sqrt{3})$.

Question 3:

We use the double angle formula $\cos 2\theta = 2 \cos^2 \theta - 1$.

$$\cos \theta = x - 2$$

Substitute this into the equation for y :

$$y = 1 + \cos 2\theta$$

$$y = 1 + (2 \cos^2 \theta - 1)$$

$$y = 2 \cos^2 \theta$$

Now substitute $\cos \theta = x - 2$:

$$y = 2(x - 2)^2$$

Thus, the Cartesian equation is $y = 2(x - 2)^2$.

Question 4:

We eliminate the parameter t by expressing t^2 in terms of y .

$$y = t^2 \implies t^2 = y$$

Substitute this into the equation for x :

$$\begin{aligned}x &= t^4 \\x &= (t^2)^2 \\x &= y^2\end{aligned}$$

Given $t \geq 0$, we analyze the restrictions:

$$\text{Since } t \geq 0, \text{ then } y = t^2 \geq 0 \text{ and } x = t^4 \geq 0.$$

The Cartesian equation is $x = y^2$ for $y \geq 0$ and $x \geq 0$.

Question 5:

We eliminate the parameter t by solving for t in the second equation.

$$\begin{aligned}y &= t - 3 \\ \implies t &= y + 3\end{aligned}$$

Substitute this into the equation for x :

$$\begin{aligned}x &= 2t^2 - 1 \\ x &= 2(y + 3)^2 - 1\end{aligned}$$

Expanding the expression:

$$\begin{aligned}x &= 2(y^2 + 6y + 9) - 1 \\ x &= 2y^2 + 12y + 18 - 1 \\ x &= 2y^2 + 12y + 17\end{aligned}$$

Thus, the Cartesian equation is $x = 2y^2 + 12y + 17$.

Question 6:

We use the identity $\sin^2 \theta + \cos^2 \theta = 1$.

$$\begin{aligned}\sin^2 \theta &= \frac{x}{4} \\ \cos \theta &= \frac{y}{4}\end{aligned}$$

Substituting these into the identity:

$$\begin{aligned}\frac{x}{4} + \left(\frac{y}{4}\right)^2 &= 1 \\ \frac{x}{4} + \frac{y^2}{16} &= 1\end{aligned}$$

Multiply by 16 to clear fractions:

$$\begin{aligned}4x + y^2 &= 16 \\ 4x &= 16 - y^2 \\ x &= 4 - \frac{y^2}{4}\end{aligned}$$

Domain of the function: Since $-1 \leq \cos \theta \leq 1$, then $-4 \leq y \leq 4$. Since $0 \leq \sin^2 \theta \leq 1$, then $0 \leq x \leq 4$. The domain of x is $[0, 4]$.

Question 7:

We isolate the trigonometric terms in the parametric equations.

$$\begin{aligned}\cos \theta &= \frac{x+1}{5} \\ \sin \theta &= \frac{y-4}{5}\end{aligned}$$

Using the identity $\sin^2 \theta + \cos^2 \theta = 1$:

$$\begin{aligned}\left(\frac{x+1}{5}\right)^2 + \left(\frac{y-4}{5}\right)^2 &= 1 \\ \frac{(x+1)^2}{25} + \frac{(y-4)^2}{25} &= 1 \\ (x+1)^2 + (y-4)^2 &= 25\end{aligned}$$

This is the equation of a circle with centre $(-1, 4)$ and radius $\sqrt{25} = 5$.

Question 8:

The curve intersects the x -axis when $y = 0$.

$$\begin{aligned}y &= t^2 - 3t \\ 0 &= t(t - 3)\end{aligned}$$

Solving for t :

$$t = 0 \quad \text{or} \quad t = 3$$

Thus, the curve intersects the x -axis at $t = 0$ and $t = 3$.

Question 9:

We are given the substitution $x = 2 - t$.

Substitute $x = 2 - t$ into $y = 3x^2 - 2x + 1$:

$$\begin{aligned}y &= 3(2-t)^2 - 2(2-t) + 1 \\ y &= 3(4 - 4t + t^2) - 4 + 2t + 1 \\ y &= 12 - 12t + 3t^2 - 3 + 2t \\ y &= 3t^2 - 10t + 9\end{aligned}$$

Thus, the parametric form is $x = 2 - t, y = 3t^2 - 10t + 9$.

Easier Exam Questions

Question 1:

We eliminate the parameter p from the equations $x = 4p - 6$ and $y = 3p$.

$$\text{From } y = 3p, \text{ we have } p = \frac{y}{3}$$

Substitute this into the equation for x :

$$\begin{aligned}x &= 4\left(\frac{y}{3}\right) - 6 \\ \implies x + 6 &= \frac{4y}{3} \\ \implies 3(x + 6) &= 4y \\ \implies 3x + 18 &= 4y \\ \implies y &= \frac{3}{4}x + \frac{18}{4} \\ \implies y &= \frac{3}{4}x + \frac{9}{2}\end{aligned}$$

Comparing this to the options, the correct answer is (B).

Question 2:

We eliminate the parameter t from $x = 5 - 2t$ and $y = t^2 - t$.

$$\begin{aligned}\text{From } x = 5 - 2t, \text{ we have } 2t &= 5 - x \\ \implies t &= \frac{5 - x}{2}\end{aligned}$$

Substitute this into the equation for y :

$$\begin{aligned}y &= \left(\frac{5 - x}{2}\right)^2 - \left(\frac{5 - x}{2}\right) \\ y &= \frac{25 - 10x + x^2}{4} - \frac{5 - x}{2} \\ \implies 4y &= 25 - 10x + x^2 - 2(5 - x) \\ \implies 4y &= 25 - 10x + x^2 - 10 + 2x \\ \implies 4y &= x^2 - 8x + 15\end{aligned}$$

Thus, the Cartesian equation is $y = \frac{1}{4}x^2 - 2x + \frac{15}{4}$.

Question 3:

We eliminate the parameter t from $x = t + 2$ and $y = 2t + 1$.

$$\text{From } x = t + 2, \text{ we have } t = x - 2$$

Substitute this into the equation for y :

$$y = 2(x - 2) + 1$$

$$y = 2x - 4 + 1$$

$$y = 2x - 3$$

Rearranging into the form $Ax + By + C = 0$:

$$2x - y - 3 = 0$$

Comparing this to the options, the correct answer is (D).

Question 4:

We eliminate the parameter t from $x = t - 2$ and $y = 2t^2 - 1$.

From $x = t - 2$, we have $t = x + 2$

Substitute this into the equation for y :

$$y = 2(x + 2)^2 - 1$$

$$y = 2(x^2 + 4x + 4) - 1$$

$$y = 2x^2 + 8x + 8 - 1$$

$$y = 2x^2 + 8x + 7$$

Comparing this to the options, the correct answer is (D).

Question 5:

We eliminate the parameter t from $x = 2t^2$ and $y = 4 - t$.

From $y = 4 - t$, we have $t = 4 - y$

Substitute this into the equation for x :

$$x = 2(4 - y)^2$$

Comparing this to the options, the correct answer is (D).

Question 6:

We eliminate the parameter t from $x = 4 - t$ and $y = (1 - t)^2$.

From $x = 4 - t$, we have $t = 4 - x$

Substitute this into the equation for y :

$$y = (1 - (4 - x))^2$$

$$y = (1 - 4 + x)^2$$

$$y = (x - 3)^2$$

Expanding the expression:

$$y = x^2 - 6x + 9$$

Thus, the Cartesian equation is $y = (x - 3)^2$ or $y = x^2 - 6x + 9$.

Question 7:

First, find the parameter value t_P for point $P(10, -3)$ using the parametric equations $x = 3t^2 - 2$ and $y = 2t + 1$.

$$\begin{aligned}\text{Using } y = 2t + 1 : \quad -3 &= 2t + 1 \\ \implies 2t &= -4 \\ \implies t_P &= -2\end{aligned}$$

Check with x : $x = 3(-2)^2 - 2 = 3(4) - 2 = 10$. (Consistent)

We are told the parameter value for Q is three less than for P :

$$\begin{aligned}t_Q &= t_P - 3 \\ t_Q &= -2 - 3 = -5\end{aligned}$$

Now find the coordinates of Q by substituting $t_Q = -5$ into the parametric equations:

$$\begin{aligned}x_Q &= 3(-5)^2 - 2 = 3(25) - 2 = 75 - 2 = 73 \\ y_Q &= 2(-5) + 1 = -10 + 1 = -9\end{aligned}$$

Therefore, the coordinates of point Q are $(73, -9)$.

Question 8:

We are looking for parametric equations that satisfy $4x^2 + y^2 = 24x - 2y - 33$.

First, complete the square for x and y :

$$\begin{aligned}4x^2 - 24x + y^2 + 2y &= -33 \\ 4(x^2 - 6x) + (y^2 + 2y) &= -33 \\ 4(x^2 - 6x + 9) + (y^2 + 2y + 1) &= -33 + 4(9) + 1 \\ 4(x - 3)^2 + (y + 1)^2 &= -33 + 36 + 1 \\ 4(x - 3)^2 + (y + 1)^2 &= 4\end{aligned}$$

Divide by 4 to put it in standard form:

$$(x - 3)^2 + \frac{(y + 1)^2}{4} = 1$$

This is an ellipse. The parametric form of an ellipse $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$ is $x = h + a \cos t$ and $y = k + b \sin t$.

$$h = 3, \quad a = 1, \quad k = -1, \quad b = 2$$

Thus, the parametric equations are:

$$x = 3 + \cos t$$

$$y = -1 + 2 \sin t$$

Comparing this to the options, (B) $x = 3 + \sin t, y = -1 + 2 \cos t$ is a valid alternative parametrization (swapping sin and cos just changes the starting point and direction of the curve).
Correct answer is (B).

Harder Exam Questions

Question 1:

Part (i): We are given $x = 4t^2$ and $y = 8t$.

To draw the curve, we find the Cartesian equation first:

$$t = \frac{y}{8}$$

$$x = 4 \left(\frac{y}{8} \right)^2 = 4 \left(\frac{y^2}{64} \right) = \frac{y^2}{16}$$

This is a parabola opening to the right. We calculate points for $-3 \leq t \leq 3$:

$$t = -3 \implies x = 4(-3)^2 = 36, \quad y = 8(-3) = -24 \quad (\text{End point})$$

$$t = 0 \implies x = 0, \quad y = 0 \quad (\text{Intercept})$$

$$t = 3 \implies x = 4(3)^2 = 36, \quad y = 8(3) = 24 \quad (\text{End point})$$

Part (ii): Find t where the parabola intersects $y = 6 - 2x$.

Substitute $x = 4t^2, y = 8t$ into $y = 6 - 2x$:

$$8t = 6 - 2(4t^2)$$

$$8t = 6 - 8t^2$$

$$8t^2 + 8t - 6 = 0$$

$$4t^2 + 4t - 3 = 0$$

Solving the quadratic equation using the quadratic formula:

$$t = \frac{-4 \pm \sqrt{4^2 - 4(4)(-3)}}{2(4)}$$

$$t = \frac{-4 \pm \sqrt{16 + 48}}{8}$$

$$t = \frac{-4 \pm \sqrt{64}}{8}$$

$$t = \frac{-4 \pm 8}{8}$$

Calculating the two values of t :

$$t_1 = \frac{4}{8} = \frac{1}{2}$$

$$t_2 = \frac{-12}{8} = -\frac{3}{2}$$

Thus, the values of t are $t = \frac{1}{2}$ and $t = -\frac{3}{2}$.

Question 2:

Part (i): We are given $x = 8t$ and $y = 4t^2$.

To draw the curve, we find the Cartesian equation:

$$t = \frac{x}{8}$$

$$y = 4\left(\frac{x}{8}\right)^2 = 4\left(\frac{x^2}{64}\right) = \frac{x^2}{16}$$

This is a parabola opening upwards. We calculate points for $-3 \leq t \leq 3$:

$$t = -3 \implies x = 8(-3) = -24, \quad y = 4(-3)^2 = 36 \quad (\text{End point})$$

$$t = 0 \implies x = 0, \quad y = 0 \quad (\text{Intercept})$$

$$t = 3 \implies x = 8(3) = 24, \quad y = 4(3)^2 = 36 \quad (\text{End point})$$

Part (ii): Find t where the curve intersects $y = 10 - \frac{3}{4}x$.

Substitute $x = 8t, y = 4t^2$ into $y = 10 - \frac{3}{4}x$:

$$4t^2 = 10 - \frac{3}{4}(8t)$$

$$4t^2 = 10 - 6t$$

$$4t^2 + 6t - 10 = 0$$

$$2t^2 + 3t - 5 = 0$$

Solving the quadratic equation by factoring:

$$(2t + 5)(t - 1) = 0$$

Thus, the values of t are $t = 1$ and $t = -\frac{5}{2}$.

Question 3:

Part (i): We are given the equation $x^2 + 4x + y^2 - 5y = 20$.

Complete the square for x and y :

$$(x^2 + 4x + 4) + (y^2 - 5y + 6.25) = 20 + 4 + 6.25$$

$$(x + 2)^2 + (y - 2.5)^2 = 30.25$$

This is a circle with centre $(-2, 2.5)$ and radius $r = \sqrt{30.25} = 5.5$.

Since the centre of the circle is where Elena is standing, Elena is at $(-2, 2.5)$.

Part (ii): The parametric equations of a circle with centre (h, k) and radius r are $x = h + r \cos \theta$ and $y = k + r \sin \theta$.

$$h = -2, \quad k = 2.5 = \frac{5}{2}, \quad r = 5.5 = \frac{11}{2}$$

Substituting these values:

$$x = -2 + \frac{11}{2} \cos \theta = \frac{-4 + 11 \cos \theta}{2}$$

$$y = \frac{5}{2} + \frac{11}{2} \sin \theta = \frac{5 + 11 \sin \theta}{2}$$

Part (iii): Find the position when $\theta = 135^\circ$.

$$\cos 135^\circ = -\frac{\sqrt{2}}{2}, \quad \sin 135^\circ = \frac{\sqrt{2}}{2}$$

Substituting these into the parametric equations:

$$x = \frac{11(-\frac{\sqrt{2}}{2}) - 4}{2} = \frac{-\frac{11\sqrt{2}}{2} - 4}{2} = \frac{-11\sqrt{2} - 8}{4}$$

$$y = \frac{11(\frac{\sqrt{2}}{2}) + 5}{2} = \frac{\frac{11\sqrt{2}}{2} + 5}{2} = \frac{11\sqrt{2} + 10}{4}$$

Therefore, the position is $\left(\frac{-11\sqrt{2}-8}{4}, \frac{11\sqrt{2}+10}{4}\right)$.

Question 4:

We are given $y = 2x^2 - 5x + 3$ and $x = 5 - 3m$.

Substitute the expression for x into the equation for y :

$$y = 2(5 - 3m)^2 - 5(5 - 3m) + 3$$

$$y = 2(25 - 30m + 9m^2) - 25 + 15m + 3$$

$$y = 50 - 60m + 18m^2 - 22 + 15m$$

$$y = 18m^2 - 45m + 28$$

Comparing this to the options, the correct answer is (B).

Question 5:

Part (i): We are given $x = 1 + \cos \theta$ and $y = \cos 2\theta$.

Use the double angle formula $\cos 2\theta = 2 \cos^2 \theta - 1$.

$$\cos \theta = x - 1$$

Substitute this into the equation for y :

$$y = 2 \cos^2 \theta - 1$$

$$y = 2(x - 1)^2 - 1$$

Expanding the expression:

$$\begin{aligned}y &= 2(x^2 - 2x + 1) - 1 \\y &= 2x^2 - 4x + 2 - 1 \\y &= 2x^2 - 4x + 1\end{aligned}$$

Part (iii): The domain of the function is determined by the range of $x = 1 + \cos \theta$.

$$\begin{aligned}-1 &\leq \cos \theta \leq 1 \\0 &\leq 1 + \cos \theta \leq 2\end{aligned}$$

Thus, the domain is $0 \leq x \leq 2$.

Question 6:

We are given $x = \cos \theta + \sin \theta$ and $y = \cos \theta - \sin \theta$.

To eliminate θ , we square both equations:

$$\begin{aligned}x^2 &= (\cos \theta + \sin \theta)^2 = \cos^2 \theta + 2 \sin \theta \cos \theta + \sin^2 \theta \\y^2 &= (\cos \theta - \sin \theta)^2 = \cos^2 \theta - 2 \sin \theta \cos \theta + \sin^2 \theta\end{aligned}$$

Using the identity $\sin^2 \theta + \cos^2 \theta = 1$:

$$\begin{aligned}x^2 &= 1 + 2 \sin \theta \cos \theta \\y^2 &= 1 - 2 \sin \theta \cos \theta\end{aligned}$$

Adding the two equations together:

$$\begin{aligned}x^2 + y^2 &= (1 + 2 \sin \theta \cos \theta) + (1 - 2 \sin \theta \cos \theta) \\x^2 + y^2 &= 2\end{aligned}$$

Thus, the Cartesian equation is $x^2 + y^2 = 2$.

Question 7:

Part (i): We are given $x = 4 \sin t + 3$ and $y = 4 \cos t - 1$.

Isolate the trigonometric terms:

$$\begin{aligned}\sin t &= \frac{x - 3}{4} \\\cos t &= \frac{y + 1}{4}\end{aligned}$$

Using the identity $\sin^2 t + \cos^2 t = 1$:

$$\begin{aligned}\left(\frac{x - 3}{4}\right)^2 + \left(\frac{y + 1}{4}\right)^2 &= 1 \\\frac{(x - 3)^2}{16} + \frac{(y + 1)^2}{16} &= 1 \\(x - 3)^2 + (y + 1)^2 &= 16\end{aligned}$$

This is the equation of a circle with centre $(3, -1)$ and radius 4. The Cartesian equation is $(x - 3)^2 + (y + 1)^2 = 16$.

Part (ii): Find the coordinates when $t = \frac{\pi}{4}$.

$$x = 4 \sin \left(\frac{\pi}{4} \right) + 3 = 4 \left(\frac{\sqrt{2}}{2} \right) + 3 = 2\sqrt{2} + 3$$

$$y = 4 \cos \left(\frac{\pi}{4} \right) - 1 = 4 \left(\frac{\sqrt{2}}{2} \right) - 1 = 2\sqrt{2} - 1$$

Therefore, the exact coordinates are $(3 + 2\sqrt{2}, -1 + 2\sqrt{2})$.

Question 8:

Part (i): We are given $x = 2 \sin \theta$ and $y = 3 + \cos 2\theta$.

Use the double angle formula $\cos 2\theta = 1 - 2 \sin^2 \theta$.

$$\sin \theta = \frac{x}{2}$$

Substitute this into the equation for y :

$$y = 3 + (1 - 2 \sin^2 \theta)$$

$$y = 4 - 2 \sin^2 \theta$$

Now substitute $\sin \theta = \frac{x}{2}$:

$$y = 4 - 2 \left(\frac{x}{2} \right)^2$$

$$y = 4 - 2 \left(\frac{x^2}{4} \right)$$

$$y = 4 - \frac{x^2}{2}$$

Part (ii): The equation $y = 4 - \frac{x^2}{2}$ is a quadratic equation in x , which represents a parabola opening downwards.

Thus, the type of graph is a parabola.

Question 9:

We are given $x = -3 + 2 \cos^2 \theta$ and $y = -5 + \cos 2\theta$.

Use the double angle formula $\cos 2\theta = 2 \cos^2 \theta - 1$.

$$\cos^2 \theta = \frac{x + 3}{2}$$

Substitute this into the equation for y :

$$y = -5 + (2 \cos^2 \theta - 1)$$

$$y = 2 \cos^2 \theta - 6$$

Now substitute $\cos^2 \theta = \frac{x+3}{2}$:

$$\begin{aligned}y &= 2 \left(\frac{x+3}{2} \right) - 6 \\y &= x + 3 - 6 \\y &= x - 3\end{aligned}$$

Restrictions: Since $0 \leq \cos^2 \theta \leq 1$, then $0 \leq \frac{x+3}{2} \leq 1$.

$$\begin{aligned}0 &\leq x + 3 \leq 2 \\-3 &\leq x \leq -1\end{aligned}$$

Thus, the Cartesian equation is $y = x - 3$ for $-3 \leq x \leq -1$.

Question 10:

The diagram describes an upper semicircular arc of radius 2, centred at $(2, -1)$.

For a circle with centre (h, k) and radius r , the parametric equations are $x = h + r \cos \theta$ and $y = k + r \sin \theta$.

$$h = 2, \quad k = -1, \quad r = 2$$

Substituting these values:

$$\begin{aligned}x &= 2 + 2 \cos \theta \\y &= -1 + 2 \sin \theta\end{aligned}$$

For an upper semicircular arc, the y -values must be greater than or equal to the y -coordinate of the centre ($y \geq -1$). This occurs when $\sin \theta \geq 0$, which corresponds to $0 \leq \theta \leq \pi$.

Comparing this to the options, the correct answer is (A).